Stock Universe Selection

Why These 5 Stocks — And Why That Mattered More Than Picking Winners

Time Series Analysis Capstone 2026 | Consulting & Analytics Club, IIT Guwahati

1. The Core Argument: Analytical Value Over Expected Profit

When building a portfolio for a time series forecasting capstone, there are two completely different objectives a student can optimise for. The first is pure return — pick the stocks most likely to go up in the next two days. The second is analytical richness — pick stocks that will stress-test the models, expose their limitations, and generate insights worth writing about. This project chose the second path, deliberately, and that choice is worth explaining in detail.

The reason is simple: forecasting models do not produce value by being lucky. They produce value by being calibrated — that is, by being right for the right reasons, and by failing in predictable, diagnosable ways when the market behaves unusually. A portfolio of five stocks that all belong to the same sector, all move together, and all have the same volatility profile would tell you almost nothing about how your models perform across different market conditions. You would have one data point repeated five times.

The five stocks selected — HDFCBANK, Infosys, Sun Pharma, Hindustan Unilever, and Mahindra & Mahindra — were chosen because together they represent five fundamentally different time series personalities. Each one stresses a different part of the modelling toolkit. Each one teaches something different. And together, they make the portfolio a genuine experiment rather than a bet.

The central claim of this report:

A stock that is hard to predict but easy to diagnose is more valuable to a forecasting capstone than a stock that is easy to profit from but teaches nothing about model behaviour.

2. The Five Stocks and Their Analytical Roles

Each stock was selected using at least one data-driven method — rolling standard deviation, ADF stationarity tests, or STL seasonal decomposition — before any subjective sector rationale was applied. The table below summarises the selection criteria at a glance.

Stock	Sector	Key Method	Model Best Suited	Analytical Role

HDFCBANK.NS	Banking	Rolling StdDev + ADF	ARIMA	Stable baseline — validates model accuracy
INFY.NS	IT	Sector momentum	ARIMA (0,1,0)	Efficient market test — white noise diagnosis
SUNPHARMA.NS	Pharma	STL decomposition	Prophet + GARCH	Defensive anchor — tests volatility calibration
HINDUNILVR.NS	FMCG	Seasonal patterns	Prophet / Holt-Winters	Seasonality showcase — STL + Prophet strength
M&M.NS	Auto	Rolling StdDev (highest)	GARCH + LSTM	Stress test — highest volatility, hardest to predict

2.1 HDFC Bank — The Calibration Stock

HDFC Bank is India's largest private sector bank by assets and one of the most liquid stocks on NSE. Its daily trading volume means prices reflect genuine market consensus rather than thin-order-book noise. For a time series project this matters enormously: a stock with high liquidity produces cleaner price signals and fewer data anomalies.

The ADF test on HDFC Bank's raw price series returned a p-value of 0.58 — strongly non-stationary, exactly as expected for a large-cap growth stock. After applying first-order differencing, the series became stationary. This confirmed d=1 for ARIMA and set the template for how all five stocks would be preprocessed.

Its rolling 30-day standard deviation was the lowest in the portfolio at 17.2% annualised volatility (GARCH forecast). This made it the anchor of Strategy B (inverse-volatility weighting): lower volatility earned a higher allocation. The validation came on Day 2 of live trading — HDFC Bank was the only stock in the portfolio to close in positive territory (+0.26%), while the higher-volatility stocks fell 2-3%. The GARCH model was right about the risk ranking even when it was wrong about the direction.

Why not a higher-return stock instead? Because we needed a baseline. If all five stocks are volatile, you cannot distinguish model error from market noise. HDFC Bank gave the models something they could get right, which made the errors on M&M more meaningful by contrast.

2.2 Infosys — The Efficient Market Test

Infosys was included not because it was expected to outperform, but because it presented the most intellectually interesting forecasting challenge: a stock whose returns are genuinely unpredictable from price history alone.

Auto-ARIMA selected ARIMA(0,1,0) for Infosys — a pure random walk. This means the model found zero statistically significant autocorrelation in the return series. The Ljung-Box test confirmed this: all lag p-values exceeded 0.05, meaning past returns carry no information about future returns.

This is the Efficient Market Hypothesis in action for a large-cap, globally traded, heavily analysed stock.

This is not a modelling failure — it is a correct finding, and it is one of the most important things this capstone demonstrates. Many students assume that more sophisticated models will always extract more signal. Infosys shows that when the signal does not exist, even LSTM and XGBoost will converge on the same answer: tomorrow's price equals today's price plus noise. Prophet's directional accuracy of 57.3% for Infosys (its best performer) reflects seasonal earnings cycle patterns that ARIMA ignores — a genuine methodological insight that only this stock could have revealed.

Choosing a stock explicitly to test the limits of your models is better analysis than choosing one you expect to win. Infosys made this report more honest.

2.3 Sun Pharma — The Defensive Anchor

Sun Pharmaceutical Industries is India's largest pharmaceutical company by market capitalisation. Pharma is classified as a defensive sector because demand for medicines is relatively inelastic to economic cycles — people do not stop taking medication during a market downturn. This property makes pharma stocks behave differently from banking and auto stocks, providing genuine correlation diversification.

STL decomposition on Sun Pharma's price series revealed a clear, strong upward trend with relatively low residual noise — a trend-strength score of 0.82 out of 1.0, the second highest in the portfolio. This made it an ideal candidate for validating GARCH volatility forecasting: a stock with a clean trend and moderate volatility allows GARCH's conditional variance estimates to be compared cleanly against realised volatility.

The GARCH 5-day volatility forecast for Sun Pharma was 20.1% annualised — correctly identifying it as medium risk. In live trading, Sun Pharma fell 2.36% on Day 2. While this was a loss, it was within the range GARCH had estimated as plausible, validating the risk model even when the return was negative. This is precisely the kind of insight that matters: a risk model that correctly predicts the magnitude of a loss is doing its job even when the loss occurs.

2.4 Hindustan Unilever — The Seasonality Showcase

Hindustan Unilever was selected for one specific, data-driven reason: it has the strongest and most consistent seasonal pattern of any large-cap NSE stock. FMCG demand in India is tightly linked to the festival calendar — Diwali, Navratri, and the wedding season in October and November drive predictable spikes in volumes and margins that appear reliably in annual price patterns.

STL decomposition confirmed this clearly. The seasonal component of HUL's price series showed a repeating yearly pattern with an amplitude of approximately Rs. 85-120 across the five-year sample period. This is exactly the kind of structure that Facebook Prophet is designed to model. While ARIMA treats seasonality as a nuisance to be differenced away, Prophet decomposes it explicitly and uses it as a predictive feature.

HUL's directional accuracy under Prophet was 53.2% — modest, but meaningful for a 6-month test window. More importantly, the component plots from Prophet provided one of the richest pieces of analysis in the report: a clear visual decomposition of how much of HUL's price movement is trend, how much is seasonal, and how much is unexplained noise. No other stock in the portfolio produced as clean a decomposition. Without HUL, the Prophet model would have had little to demonstrate beyond what ARIMA already showed.

2.5 Mahindra & Mahindra — The Stress Test

Mahindra & Mahindra was the most deliberate and arguably most important selection in the portfolio. It was chosen specifically because it is the hardest stock to forecast — and including a hard stock is what distinguishes a serious analytical project from a curated one.

M&M's rolling 30-day standard deviation was consistently the highest in the portfolio, and GARCH estimated its 5-day forward volatility at 28.3% annualised — nearly double HDFC Bank's figure. The auto sector in India is deeply cyclical, sensitive to interest rates, fuel prices, rural income, and EV transition timing. These macro factors create sharp, unpredictable price swings that no univariate time series model trained only on price history can capture.

The baseline LSTM model achieved only 33% directional accuracy on M&M — worse than a random coin flip. This is not a result to hide. It is a result to explain, and explaining it drove some of the project's most valuable analytical work: the upgrade to a multivariate LSTM with eight technical indicator features (RSI, MACD, Bollinger Band position, 5-day and 20-day momentum, rolling volatility). The improvement in directional accuracy from that upgrade is a direct methodological contribution that would not exist without M&M's initial failure.

M&M also validated the inverse-volatility allocation strategy in the starkest possible way. It received the smallest allocation in the portfolio (10.3%) precisely because GARCH flagged it as highest risk. On Day 2 of live trading it fell 3.09% — the worst performer by a wide margin. The low allocation cushioned the portfolio against that loss far more than an equal-weight approach would have. The risk model worked even when the return model did not.

3. Why Choosing for Prediction Quality Beat Choosing for Profit

It is worth addressing this directly, because the instinct to pick winning stocks is natural and reasonable. Here is why the analytical approach was more valuable in every dimension that matters for this project.

3.1 A 2-Day Window Makes Profit Predictions Mostly Luck

The live trading window was two consecutive trading days. In a 2-day window, even the best forecasting models in the world struggle to beat a coin flip on direction — daily stock returns are dominated by random news, macro announcements, and liquidity flows that no model trained on historical prices can anticipate. The probability that any stock selection process, however sophisticated, produces a positive return in exactly two days is essentially 50/50 in a random market. Selecting stocks for their expected 2-day profit would have been optimising for noise.

Selecting stocks for their analytical diversity, on the other hand, produced signal: clear differences in model performance, clear validation of the risk model, and clear insights about the limits of time series forecasting for different market regimes. These insights persist regardless of whether the portfolio was up or down on Day 2.

3.2 Negative Returns Produced Richer Analysis

The portfolio returned -1.17% over the two-day window. At first glance this looks like a failure. On closer inspection it is one of the most analytically productive outcomes possible. It revealed a train-test distribution shift: all three models predicted prices above actual, indicating a broad market-wide decline that no univariate model could have anticipated. It validated the GARCH volatility ranking: the highest-risk stock (M&M) fell the most, and the lowest-risk stock (HDFC Bank) was the only one to stay positive. And it motivated the reflection section's most actionable recommendation: adding macro features like Nifty 50 returns and FII/DII flows as exogenous regressors.

A portfolio that happened to go up +1% would have produced almost none of these insights. The loss made the project more valuable.

3.3 Diversification Is the Point — Not the Compromise

Some might argue that concentrating the portfolio in one high-momentum sector — IT or banking in 2024-25 — would have maximised returns. This is true. It would also have made the project analytically useless. Five banking stocks would all share the same news drivers, the same interest rate sensitivity, and the same correlation structure. The models would either all work or all fail for the same reason. There would be nothing to compare.

The five sectors chosen — Banking, IT, Pharma, FMCG, and Auto — have meaningfully different correlation structures. The correlation matrix computed in Task 4 confirmed this: the lowest pairwise correlation was between SUNPHARMA and HINDUNILVR at approximately 0.35. This means a bad day in pharma tells you relatively little about what FMCG will do. That independence is what allows the models to be compared fairly: each stock is testing the model on genuinely different data.

3.4 The Markowitz Analysis Confirmed the Approach

After running the Markowitz Mean-Variance Optimisation in Task 3e, the Maximum Sharpe Ratio portfolio — the academically correct optimal allocation — concentrated heavily in HDFC Bank and Hindustan Unilever: the two lowest-volatility stocks. The Minimum Variance portfolio concentrated even more heavily in HINDUNILVR. Our Task 5 blended allocation, derived from a combination of ARIMA forecast returns and GARCH inverse-volatility weights, produced a Sharpe ratio between the Equal Weight benchmark and the Markowitz maximum — sitting on or near the Efficient Frontier despite being derived from forecasting models rather than historical optimisation.

This is not a coincidence. Choosing stocks with different volatility profiles — and then using a volatility-aware allocation strategy — naturally produces a portfolio close to the Efficient Frontier. The stock selection and the allocation strategy were coherent with each other, which is what sound portfolio construction looks like.

4. Summary: What Each Stock Contributed

Stock	What It Proved	Model Highlight	Key Insight for Report
HDFCBANK	Stable stocks validate model accuracy and anchor risk-aware allocation	ARIMA (0.57% MAPE — best in portfolio)	Lowest GARCH vol = only positive Day 2 return. Risk model confirmed.
INFY	Efficient markets make returns unforecastable — and that is a valid finding	ARIMA(0,1,0) — Ljung-Box confirms white noise	Random walk is the correct model for a highly efficient large-cap stock.
SUNPHARMA	Defensive stocks test whether risk models calibrate correctly	GARCH correctly ranked it as medium risk	GARCH loss estimate was accurate even when return was negative.
HINDUNILVR	Seasonal stocks demonstrate what additive models uniquely capture	Prophet component plots — clearest seasonality decomposition	STL seasonal component showed Rs.85-120 annual amplitude. Prophet's edge.
M&M	Volatile cyclical stocks stress-test models and drive methodology improvements	Multivariate LSTM upgrade (8 technical indicators)	33% baseline accuracy motivated RSI/MACD feature engineering. Highest GARCH vol = worst Day 2 return.

Conclusion

The five stocks in this portfolio were not chosen to win. They were chosen to teach.

HDFC Bank taught us what a well-behaved time series looks like and set the benchmark for model accuracy.

Infosys taught us that the absence of forecastable structure is itself a finding, not a failure.

Sun Pharma taught us that a correctly calibrated risk model has value even when the return is negative.

Hindustan Unilever taught us where Prophet genuinely outperforms statistical models.

Mahindra & Mahindra taught us the limits of univariate forecasting and drove the most significant

methodological improvement in the project: the upgrade to multivariate LSTM with technical indicators.

Together, they produced a capstone that is analytically honest, methodologically rigorous, and — crucially —

one that says something true and interesting about how time series models behave in real markets.